
FLIGHTEC
PRECISE DATA FOR
MINING AND MINERAL EXPLORATION
Above.
Below.
Beyond.
Flightec delivers precise, high-quality geophysical survey data that gives exploration teams greater confidence in where to target next - reducing uncertainty and managing risk.
flightec.co.za

FLIGHTEC

Turning Data into insight unlocks potential.

Flightec is Southern Africa's specialist partner in mining and mineral exploration. We bring together technology development, engineering, field operations and data processing to help exploration teams understand the area they're exploring.

From airborne surveys to measurements captured above and below the surface, we connect insights from above and below the surface to build a clearer picture of each target.

Our purpose driven teams engineer, develop, and deploy advanced technology that supports better-informed exploration decisions.

Our vision is to make mining and mineral exploration more accessible across Southern Africa, even in the most challenging environments.

Making the inaccessible, accessible.

Remote locations and complex terrain can limit how exploration teams investigate an area. Our unmanned aircraft systems extend survey access across rugged and difficult-to-reach environments, including locations where conventional manned aircraft face operational constraints.

With fewer mobilisation and operating demands, unmanned surveys can also offer a more cost-effective solution for suitable projects.

Tomorrow's technology, today.

We have developed advanced unmanned aircraft systems and sensors, purpose-built for the challenges of mining and mineral exploration.

Collaboration between our engineers and field operations teams, working together with geophysicists to support projects through practical technology — designing, testing and refining systems to capture the data our clients need.

FLIGHTEC

Above.

Greater Detail. A Clearer Direction.

Our airborne geophysical services capture information about the terrain and geology across your project area.

TOPOGRAPHIC SURVEY

Understand the terrain at the level where decisions are made.

Flightec captures high-resolution topographic data to support terrain analysis, access planning, survey design and site development.

Detailed orthophotos, digital terrain models and digital surface models help teams understand elevation, surface features and the practical conditions that shape a field programme.

MAGNETIC SURVEY

See what greater magnetic resolution can reveal.

Flightec captures high-resolution, low-altitude magnetic data to support structural mapping, target definition and exploration planning.

Greater spatial detail helps teams examine magnetic patterns and anomalies, refine geological interpretations and identify areas for further investigation.

RADIOMETRIC SURVEY

Explore the geological picture at the surface.

Radiometric surveys measure responses associated with naturally occurring potassium, uranium and thorium near the Earth's surface.

These measurements help exploration teams examine lithological variation, geological boundaries and potential alteration patterns. Viewed together, the individual channels and total count response provide complementary perspectives on the survey area.

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Below.

Depth of Expertise.

Flightec's wireline logging services deliver continuous, depth-referenced borehole data to support geological correlation and structural interpretation. Integrated with drilling records and airborne surveys, these measurements build a clearer understanding of subsurface geology.

OPTICAL BOREHOLE IMAGING

Add continuous structural context to the drilling record.

Optical borehole imaging captures a continuous visual record of the borehole wall. It helps teams examine features such as fractures, contacts and changes in rock character, adding context to geological logging. The imagery supports structural interpretation and comparison with the core record, helping build a more detailed understanding of conditions downhole.

MAGNETIC SUSCEPTIBILITY LOGGING

Connect surface magnetic patterns with measurements at depth.

Magnetic susceptibility logging measures how strongly the rocks intersected by a borehole respond to an applied magnetic field.

The depth-referenced measurements help identify changes in magnetic properties between geological units and support interpretation of the sources contributing to airborne magnetic anomalies.

NATURAL GAMMA RAY LOGGING

Add another layer to the geological model.

Natural gamma ray logging records variations in naturally occurring gamma radiation along the borehole. These measurements help distinguish changes between geological units and support correlation between boreholes. Used alongside other logs and drilling information, they add a further reference for interpreting the subsurface geology.

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Discuss Your Project

Every survey begins with a clear exploration question. We work with your team to define the scope, match the technology to your objectives and deliver the data you need to move forward.



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Souther African Capability, Built for Regional Realities.

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Windhoek, Namibia

RESEARCH & DEVELOPMENT
Knysna, South Africa

OPERATIONS HUB
eMalahleni, South Africa

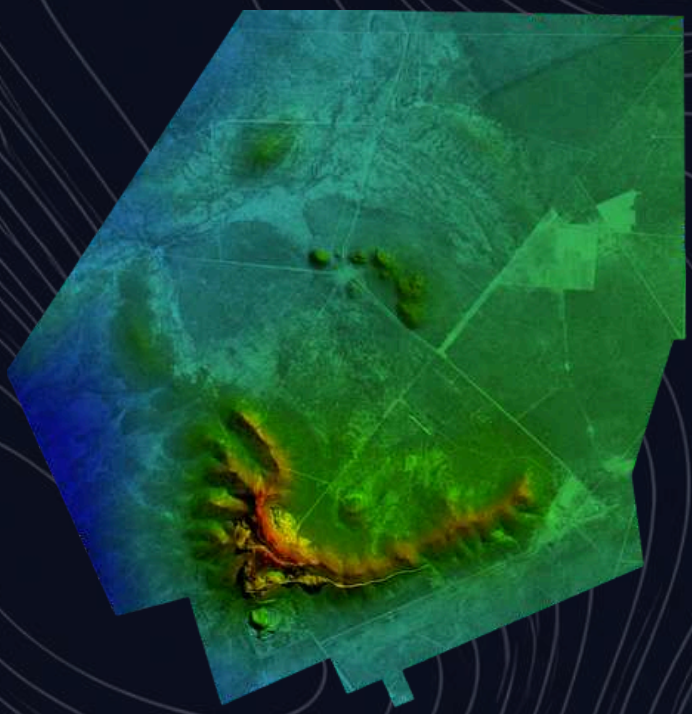
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Better decisions begin with better quality data.

Topographic Survey Comparison

The SRTM dataset provides broad terrain context. Flightec's survey reveals finer surface features, giving teams a more detailed basis for access planning and site assessment.

01 Flightec Survey



Flightec Topographic Survey

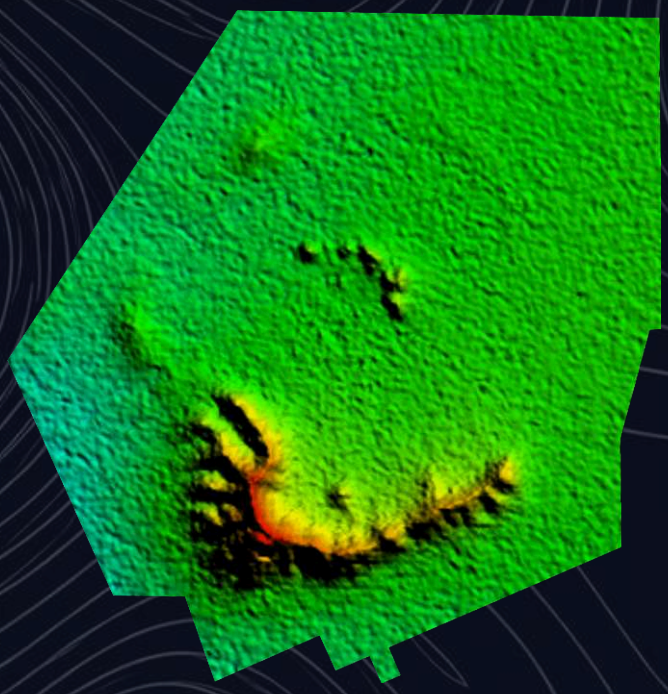
Area Coverage:

10,000 ha

Captured GSD:

5 cm

02 SRTM



Shuttle Radar Topography Mission

Area Coverage:

10,000 ha

Captured GSD 1 Arc-Second:

~30 m

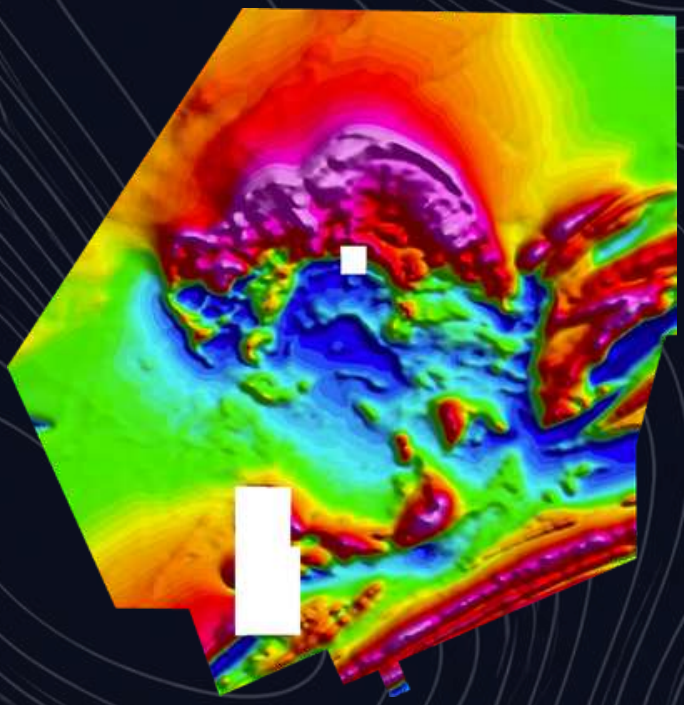
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Magnetic Survey Comparison

The regional survey shows the broader magnetic patterns. Flightec's dataset resolves finer variations within these anomalies, supporting more detailed structural interpretation and target evaluation.

01 Flightec Magnetic Survey



Area Coverage: 10,000 ha

Flight Line Spacing: 50 m

Tie Line Spacing: 500 m

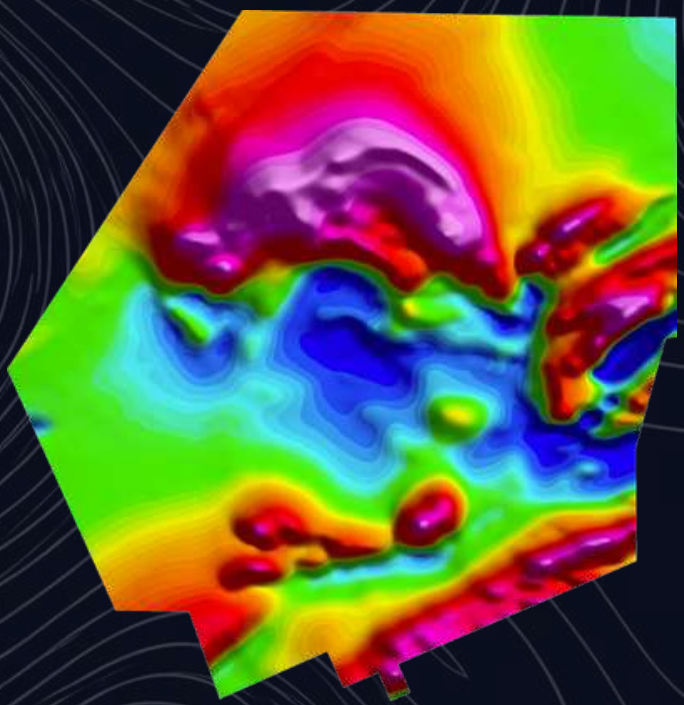
Flight Line Direction: (UTM Grid) 0°/180°

Tie Line Direction: (UTM Grid) 90°/270°

Nominal Terrain Clearance: 30 m

Sampling Interval Magnetics: 0.001 sec

02 Regional Magnetic Survey



Area Coverage: 10,000 ha

Flight Line Spacing: 200 m

Tie Line Spacing: 2500 m

Flight Line Direction: (UTM Grid) 0°/180°

Tie Line Direction: (UTM Grid) 90°/270°

Nominal Terrain Clearance: 80 m

Sampling Interval Magnetics: 0.1 sec

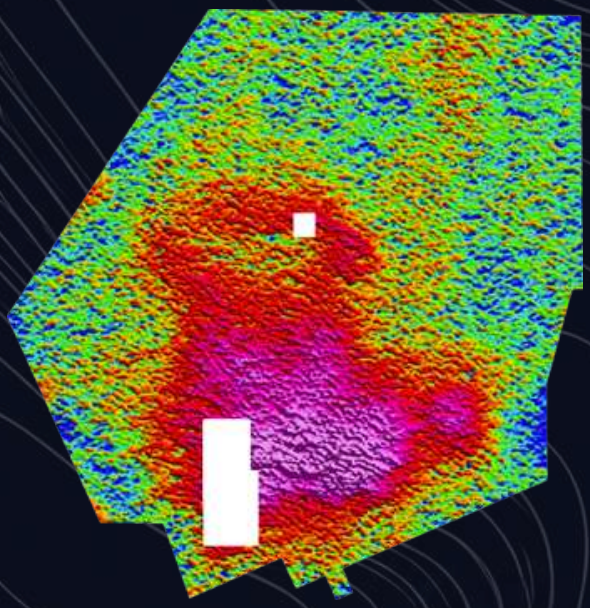
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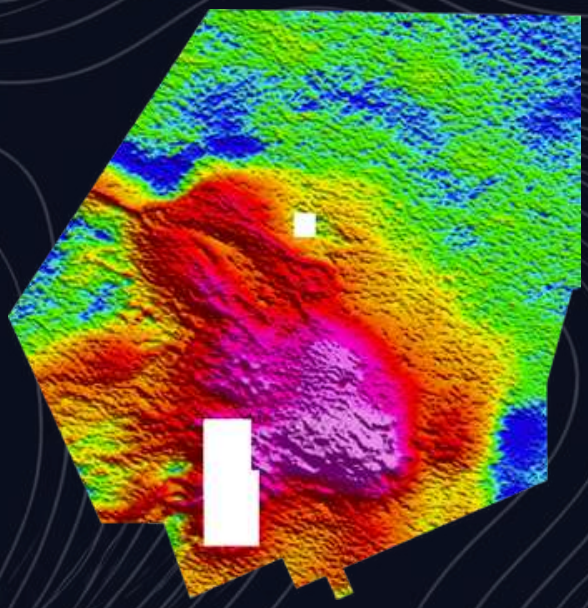
Radiometric Survey Data

The uranium, thorium, potassium and total count maps show different aspects of the same survey area. Read together, they help distinguish patterns that may reflect changes in near-surface geology.

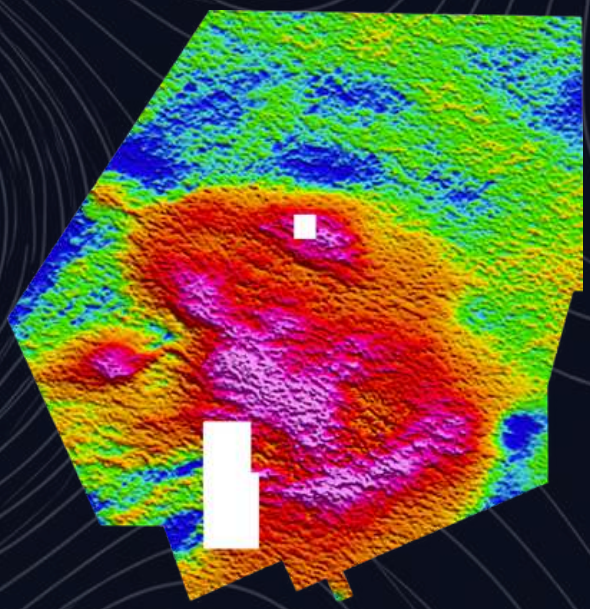
01 Uranium



02 Thorium



03 Potassium



04 Total Count

